

Linear differential equations

3.1

Linear differential equations of order n

A differential equation of order n , of the form

$$y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_{n-1}(x)y' + a_n(x)y = f(x) \quad (3.1.1)$$

where a_i , $i = 1, \dots, n$ and f are continuous functions on the interval $[a, b]$, is called **linear differential equation of order n** .

The equation (3.1.1) can be written in the form:

$$L[y] = f(x) \quad (3.1.2)$$

where $L : C^n[a, b] \rightarrow C[a, b]$ is the operator defined by

$$L[y] = y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_n(x)y.$$

If $f = 0$, the equation is called **homogeneous** and if $f \neq 0$, the equation is called **nonhomogeneous**.

We consider the homogeneous equation $L[y] = 0$. The set of solutions for this equation is a subspace of $C^n[a, b]$ and is denoted by $\text{Ker}L$:

$$\text{Ker}L = \{y \in C^n[a, b] \mid L[y] = 0\}. \quad (3.1.3)$$

It can be proved that the dimension of $\text{Ker}L$ is n . A basis for $\text{Ker}L$ is called a **fundamental system of solutions** for the homogeneous equation.

Definition 3.1.1 The system of functions $y_1, \dots, y_n \in C[a, b]$ is called **linearly dependent** on the interval $[a, b]$ if there exist the constants $\alpha_1, \dots, \alpha_n$, not all zero, such that

$$\alpha_1 y_1(x) + \alpha_2 y_2(x) + \dots + \alpha_n y_n(x) = 0, \quad \forall x \in [a, b].$$

In the other case, if this identity holds only for $\alpha_1 = \alpha_2 = \dots = \alpha_n = 0$, the system of functions is called **linearly independent**.

 **Example 3.1.1** Show that the system of functions e^x, e^{2x}, e^{3x} is linearly independent on each real interval.

 **Solution:** We assume the contrary, that is, there exist $\alpha_1, \alpha_2, \alpha_3$, not all zero, such that

$$\alpha_1 e^x + \alpha_2 e^{2x} + \alpha_3 e^{3x} = 0, \quad \forall x \in \mathbb{R}.$$

Assume that $\alpha_3 \neq 0$. Dividing the relation by e^x we obtain $\alpha_1 + \alpha_2 e^x + \alpha_3 e^{2x} = 0$. We differentiate the relation and divide by e^x . We have $\alpha_2 + 2\alpha_3 e^x = 0$, and differentiating once more $2\alpha_3 e^x = 0$. This is impossible, since we assumed $\alpha_3 \neq 0$. So the system is linearly independent. 

 **Example 3.1.2** Prove that the functions $x^2, x - 1$ and $2x^2 + 3x - 3$ are linearly dependent on each real interval.

 **Solution:** Since $2x^2 + 3(x - 1) - (2x^2 + 3x - 3) = 0$, there exist the nonzero constants $\alpha_1 = 2, \alpha_2 = 3, \alpha_3 = -1$ from the definition of the linear independence. 

Definition 3.1.2 The determinant

$$W(x, y_1, \dots, y_n) = \begin{vmatrix} y_1(x) & y_2(x) & \dots & y_n(x) \\ y_1'(x) & y_2'(x) & \dots & y_n'(x) \\ \dots & \dots & \dots & \dots \\ y_1^{(n-1)}(x) & y_2^{(n-1)}(x) & \dots & y_n^{(n-1)}(x) \end{vmatrix}$$

is called the **Wronskian** of the functions $y_1, \dots, y_n$; $y_i \in C^{n-1}[a, b]$, $i = 1, \dots, n$.

Theorem 3.1.3 If $y_1, \dots, y_n$ are linearly dependent, then $W(x, y_1, \dots, y_n) = 0$, for every $x \in [a, b]$.

The converse of this Theorem is not true. In fact, we have:

Theorem 3.1.4 If $y_1, \dots, y_n$ are linearly independent and satisfy the equation $L[y] = 0$, then $W(x, y_1, \dots, y_n) \neq 0$, for each $x \in [a, b]$.

 **Example 3.1.3** Investigate the linear dependence of the following functions on the interval $[0, \frac{\pi}{2}]$: $y_1(x) = 2$, $y_2(x) = \cos x$, $y_3(x) = \cos 2x$.

 **Solution:** We have:

$$\begin{aligned} W(x, y_1, y_2, y_3) &= \begin{vmatrix} 2 & \cos x & \cos 2x \\ 0 & -\sin x & -2 \sin 2x \\ 0 & -\cos x & -4 \cos 2x \end{vmatrix} = 8 \sin x \cos 2x - 4 \cos x \sin 2x \\ &= 8 \sin x (1 - 2 \sin^2 x) - 8 \cos^2 x \sin x = -8 \sin^3 x \neq 0 \text{ on } \left[0, \frac{\pi}{2}\right]. \end{aligned}$$

Theorem 3.1.3 implies that the system is linearly independent. 

The general solution of the linear homogeneous equation $L[y] = 0$ is

$$y = C_1 y_1 + C_2 y_2 + \cdots + C_n y_n,$$

where $y_1, y_2, \dots, y_n$ are linearly independent solutions of the equation (a fundamental system of solutions), and $C_1, C_2, \dots, C_n$ are arbitrary constants.

The general solution of the nonhomogeneous equation $L[y] = f(x)$ is of the form $y = y_0 + y_p$, where y_0 is the general solution of the homogeneous equation $L[y] = 0$ and y_p is a particular solution of the **nonhomogeneous equation** $L[y] = f(x)$.

3.2

Linear differential equations of order n , with constant coefficients

The linear differential equation of order n :

$$y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} y' + a_n y = 0, \quad (3.2.1)$$

where $a_1, \dots, a_n \in \mathbb{R}$ is called **linear and homogeneous, with constant coefficients**. For such an equation, a fundamental system of solutions can be determined.

To this purpose, we associate the **characteristic equation**

$$r^n + a_1 r^{n-1} + \cdots + a_{n-1} r + a_n = 0.$$

and determine the roots. We have:

Theorem 3.2.1 1) If r is a real, simple root of the characteristic equation, the corresponding solution of the differential equation is

$$y_1(x) = e^{rx}.$$

2) If $a \pm ib$ is a pair of complex conjugated, simple roots, then the two corresponding linearly independent solutions are

$$y_1(x) = e^{ax} \cos bx \text{ and } y_2(x) = e^{ax} \sin bx.$$

3) If r is a real root with the multiplicity order q , then the q linearly independent solutions of the differential equation are

$$y_1(x) = e^{rx}, y_2(x) = xe^{rx}, \dots, y_q(x) = x^{q-1}e^{rx}$$

4) If $a \pm ib$ is a pair of conjugated, complex roots with the multiplicity order q , then the $2q$ linearly independent solutions of the differential equation are:

$$\begin{aligned} y_1(x) &= e^{ax} \cos bx, \quad y_2(x) = xe^{ax} \cos bx, \quad \dots, \quad y_q(x) = x^{q-1}e^{ax} \cos bx \\ y_{q+1}(x) &= e^{ax} \sin bx, \quad y_{q+2}(x) = xe^{ax} \sin bx, \quad \dots, \quad y_{2q}(x) = x^{q-1}e^{ax} \sin bx. \end{aligned} \quad (3.2.2)$$

Remark 3.2.2 The number of the solutions constructed in this way is equal to the order of the equation. A fundamental system of solutions for the differential equation is obtained.

 **Example 3.2.1** Determine the general solution of the equation $y''' - 5y'' + 8y' - 6y = 0$.

 **Solution:** The characteristic equation is $r^3 - 5r^2 + 8r - 6 = 0$, with the roots $r_1 = 3$, $r_2 = 1 + i$ and $r_3 = 1 - i$.

To the root r_1 it corresponds the solution $y_1 = e^{3x}$. To the pair of complex conjugated roots correspond $y_2 = e^x \cos x$ and $y_3 = e^x \sin x$.

The general solution of the equation is $y = C_1e^{3x} + C_2e^x \cos x + C_3e^x \sin x$. 

 **Example 3.2.2** Determine the general solution of the equation $y''' + 10y'' + 25y' = 0$.

 **Solution:** The characteristic equation is $r^3 + 10r^2 + 25r = 0$, with $r_1 = 0$ a simple root and $r_{2,3} = -5$ a double one. The corresponding linearly independent solutions are $y_1 = 1$, $y_2 = e^{-5x}$ and $y_3 = xe^{-5x}$, so the general solution is $y = C_1 + C_2e^{-5x} + C_3xe^{-5x}$. 

 **Example 3.2.3** Solve the Cauchy problem $y''' + y'' = 0$, $y(0) = 1$, $y'(0) = 0$, $y''(0) = 1$.

☞ **Solution:** The general solution is $y = C_1 + C_2x + C_3e^{-x}$. Using the initial conditions we determine the constants C_1, C_2, C_3 . We have the system

$$\begin{cases} C_1 + C_3 = 1 \\ C_2 - C_3 = 0 \\ C_3 = 1 \end{cases}$$

with the solutions $C_1 = 0, C_2 = 1, C_3 = 1$, and so $y = x + e^{-x}$. ☑



Exercices

3.2.1 Solve the following equations:

i). $8y'' - 2y' - 3y = 0$

v). $y^{IX} = 0$

ii). $4y'' - 8y' + 5y = 0$

vi). $y''' - 8y = 0$

iii). $y^{IV} - 2ay'' + a^2y = 0, a > 0$

vii). $y^{VII} - y^{VI} + y^V - y^{IV} = 0$

iv). $3y^{IV} - 4y''' + y'' = 0$

viii). $y^{IV} + 2y'' + y = 0$

3.2.2 Solve the following initial value or bi-local problems:

i). $y'' - 2y' + 2y = 0, y(0) = 0, y'(0) = 1$

ii). $y'' - 5y' + 4y = 0, y(0) = 5, y'(0) = 8$

iii). $y'' + 4y = 0, y(0) = 1, y(\frac{\pi}{4}) = -1$

iv). $y'' + y' - 2y = 0, y'(0) = 2, \lim_{x \rightarrow \infty} y(x) = 0$

v). $y^{IV} - y = 0, y(0) = 1, y'(0) = 0, y(\pi) = e^\pi, y'(\pi) = 0$

Example 3.2.4 (Application) Consider a body of mass m , elasticity constant k , situated in a media of viscosity γ . The body is displaced from its equilibrium position and then is left free. Knowing that the friction force is proportional to the speed of the body, study its movement.

☞ **Solution:** Denoting the movement law by $y(t)$ and applying the second equation of Newton, we have $My'' = -ky - \gamma y'$ or

$$y'' + \frac{\gamma}{M}y' + \omega_0^2y = 0,$$

where $\omega_0 = \sqrt{\frac{k}{M}}$ is the own pulsation of the damped oscillator and $-ky$ is the elastic force. One can see that the resulting equation is a linear, homogeneous equation of the second order.

I. In the ideal case, when the friction is neglected, we have $y'' + \omega_0^2 y = 0$. The characteristic equation is $r^2 + \omega_0^2 = 0$, with the roots $r_{1,2} = \pm i\omega_0$, so $y = C_1 \sin \omega_0 t + C_2 \cos \omega_0 t$. The solution can also be written in the form $y = y_0 \cos(\omega_0 t + \varphi)$, where y_0 is called amplitude, φ is called phase, and they are determined from the initial conditions of the movement.

II. In the more realistic case when the friction is not zero ($\gamma \neq 0$), the characteristic equation is $r^2 + \frac{\gamma}{M}r + \omega_0^2 = 0$, with $\Delta = \left(\frac{\gamma}{M}\right)^2 - 4\omega_0^2$.

a) If $\Delta > 0$, we have $r_{1,2} = \frac{-\frac{\gamma}{M} \pm \sqrt{\frac{\gamma^2}{M^2} - 4\omega_0^2}}{2} < 0$, and so $y = C_1 e^{r_1 t} + C_2 e^{r_2 t}$.

The movement is not periodical, the body comes back slowly to the equilibrium position, without oscillating. It is called an overdamped motion.

b) If $\Delta < 0$, we have $r_{1,2} = \frac{-\frac{\gamma}{M} \pm i\sqrt{4\omega_0^2 - \frac{\gamma^2}{M^2}}}{2}$, from where $y = y_0 e^{-\frac{\gamma}{2M}t} \cos(\omega_0' t +$

$\varphi)$, with $\omega_0'^2 = \omega_0^2 - \frac{\gamma^2}{4M^2}$. The movement is pseudo-sinusoidal, underdamped. The body oscillates around the equilibrium position, with an amplitude that decreases exponentially towards zero.

c) If $\Delta = 0$, we have $r_1 = r_2 = -\frac{\gamma}{2M}$ and $y = C_1 e^{-\frac{\gamma}{2M}t} + C_2 t e^{-\frac{\gamma}{2M}t}$.

The damp is called critical, the body comes back to the equilibrium position faster than in the first case. 



Answers

3.2.1 i) $y = C_1 e^{\frac{3}{4}x} + C_2 e^{-\frac{1}{2}x}$; ii) $y = C_1 e^x \cos \frac{1}{2}x + C_2 e^x \sin \frac{1}{2}x$; iii) $y = (C_1 + C_2 x)e^{\sqrt{ax}} + (C_3 + C_4 x)e^{-\sqrt{ax}}$; iv) $y = C_1 + C_2 x + C_3 e^x + C_4 e^{\frac{1}{3}x}$; v) $y = C_1 + C_2 x + C_3 x^2 + \dots + C_9 x^8$; vi) $y = C_1 e^{2x} + C_2 e^{-x} \cos x \sqrt{3} + C_3 e^{-x} \sin x \sqrt{3}$; vii) $y = C_1 + C_2 x + C_3 x^2 + C_4 x^3 + C_5 e^x + C_6 \cos x + C_7 \sin x$; viii) $y = C_1 \cos x + C_2 x \cos x + C_3 \sin x + C_4 x \sin x$

3.2.2 i) $y = e^x \sin x$; ii) $y = 4e^x + e^{4x}$; iii) $y = \cos 2x - \sin 2x$; iv) $y = -e^{-2x}$; v) $y = \frac{1}{2}e^x + \frac{e^\pi}{2}e^{-x} + \frac{1}{2}(1 - e^\pi) \cos x - \frac{1}{2}(1 - e^\pi) \sin x$.

Linear, nonhomogeneous equations of order n , with constant coefficients

A linear, nonhomogeneous, differential equation of order n , with constant coefficients has the form:

$$y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} y' + a_n y = f(x), \quad (3.3.1)$$

$a_1, \dots, a_n \in \mathbb{R}$, $f \in C[a, b]$, $f \neq 0$.

Its general solution is $y = y_0 + y_p$, y_0 being the general solution of the associated homogeneous equation, and y_p a particular solution of the nonhomogeneous equation. In what follows we present two methods of determining a particular solution.

The method of Lagrange (variation of constants)

We suppose that the corresponding homogeneous equation $L[y] = 0$ has the general solution $y_0 = C_1 y_1 + C_2 y_2 + \cdots + C_n y_n$. The particular solution of the equation $L[y] = f(x)$ is searched in the form $y_p = \varphi_1(x) y_1 + \varphi_2(x) y_2 + \cdots + \varphi_n(x) y_n$, where the functions $\varphi_1, \varphi_2, \dots, \varphi_n$ have to be deduced from the system:

$$\begin{cases} \varphi_1' y_1 + \varphi_2' y_2 + \cdots + \varphi_n' y_n = 0 \\ \varphi_1' y_1' + \varphi_2' y_2' + \cdots + \varphi_n' y_n' = 0 \\ \dots\dots\dots \\ \varphi_1' y_1^{(n-2)} + \varphi_2' y_2^{(n-2)} + \cdots + \varphi_n' y_n^{(n-2)} = 0 \\ \varphi_1' y_1^{(n-1)} + \varphi_2' y_2^{(n-1)} + \cdots + \varphi_n' y_n^{(n-1)} = f(x) \end{cases}$$

Remark 3.3.1 The method can be applied also to linear differential equations with variable coefficients, of the form (3.1.1).

 **Example 3.3.1** Determine the general solution of the equation $y'' + 2y' = \frac{2x-1}{x^2}$.

 **Solution:** We solve first the homogeneous equation $y'' + 2y' = 0$ and obtain $y_0 = C_1 + C_2 e^{-2x}$.

We look for the particular solution in the form $y_p = \varphi_1(x) + \varphi_2(x) e^{-2x}$, with φ_1, φ_2 solutions of the system

$$\begin{cases} \varphi_1' + \varphi_2' e^{-2x} = 0 \\ -2\varphi_2' e^{-2x} = \frac{2x-1}{x^2} \end{cases}$$

We get $\varphi'_2 = -e^{2x} \frac{2x-1}{2x^2}$, so $\varphi_2 = -\frac{e^{2x}}{2x}$ and $\varphi'_1 = \frac{2x-1}{2x^2}$, $\varphi_1 = \ln|x| + \frac{1}{2x}$.

Replacing the results in y_p , we have

$$y_p = \ln|x| + \frac{1}{2x} - \frac{e^{2x}}{2x} \cdot e^{-2x} = \ln|x|$$

and so $y = C_1 + C_2 e^{-2x} + \ln|x|$. ☑

 **Example 3.3.2** Integrate the equation $y'' - 2y' + 2y = \frac{e^x}{\sin 2x}$.

 **Solution:** The solution for the homogeneous equation is $y_0 = C_1 e^x \cos x + C_2 e^x \sin x$, so we search for $y_p = \varphi_1(x) e^x \cos x + \varphi_2(x) e^x \sin x$, where

$$\begin{cases} \varphi'_1 e^x \cos x + \varphi'_2 e^x \sin x = 0 \\ \varphi'_1 (e^x \cos x - e^x \sin x) + \varphi'_2 (e^x \sin x + e^x \cos x) = \frac{e^x}{\sin 2x} \end{cases}.$$

We obtain the general solution:

$$y = C_1 e^x \cos x + C_2 e^x \sin x + \frac{1}{2} e^x \cos x \ln \left| \frac{\tan \frac{x}{2} - 1}{\tan \frac{x}{2} + 1} e^x \sin x \ln \left| \tan \frac{x}{2} \right| \right|.$$

☑



Exercices

 **3.3.1** Integrate the equations:

i). $y'' + y = \frac{1}{\sin x}$

v). $y'' - y = \frac{e^{2x}}{e^x + 1}$

ii). $y'' - 2y' = e^{3x} \sin e^x$

vi). $y''' + y' = \frac{\sin x}{\cos^2 x}$

iii). $y'' - 2y' + y = \frac{e^x}{x}$

vii). $y''' + y'' = \frac{x-1}{x^2}$

iv). $y'' - y' = \frac{1}{e^x - 1}$

viii). $y'' + 2y' + y = 5e^{-x} \sqrt{x}$

The method of undetermined coefficients

In certain situations the form of a particular solution y_p can be predicted from the form of the right-hand term of the equation $L[y] = f$. We distinguish the following cases:

- 1) $f(x) = e^{\alpha x} P_m(x)$, with $P_m(x)$ a polynomial of degree m in x and $\alpha \in \mathbb{R}$.
 a) If α is not a root of the characteristic equation, then $y_p = e^{\alpha x} Q_m(x)$;
 b) If α is a multiple root of order q for the characteristic equation, then $y_p = x^q e^{\alpha x} Q_m(x)$, where $Q_m(x)$ is a polynomial of degree m whose coefficients have to be determined from the condition that y_p satisfies the equation $L[y] = f$.

2) $f(x) = e^{\alpha x} P_m(x) \cos \beta x$ or $f(x) = e^{\alpha x} P_m(x) \sin \beta x$, $\alpha, \beta \in \mathbb{R}$

- a) If $\alpha + i\beta$ is not a root of the characteristic equation, then $y_p = e^{\alpha x} [Q_m(x) \cos \beta x + R_m(x) \sin \beta x]$;
 b) If $\alpha + i\beta$ is a multiple root of order q of the characteristic equation, then $y_p = x^q e^{\alpha x} [Q_m(x) \cos \beta x + R_m(x) \sin \beta x]$, where $Q_m(x)$ and $R_m(x)$ are polynomials of degree m , whose coefficients are determined by identification from the condition that y_p satisfies the equation $L[y] = f$.

 **Example 3.3.3** Determine the general solution of the equation $y'' - 4y' = xe^{4x}$.

 **Solution:** The general solution of the homogeneous equation is $y_0 = C_1 + C_2 e^{4x}$. The right-hand term $f = xe^{4x}$ is of the form $e^{\alpha x} P_m(x)$, with a polynomial of degree 1. Since $\alpha = 4$ is a simple root of the characteristic equation, we search the particular solution in the form $y_p = x(ax + b)e^{4x}$. Using the condition that y_p satisfies the equation, we have:

$$(16ax^2 + 16ax + 16bx + 2a + 8b)e^{4x} - 4(4ax^2 + 2ax + 4bx + b)e^{4x} = xe^{4x}$$

Identifying the coefficients, we obtain the system $\begin{cases} 8a = 1 \\ 2a + 4b = 0 \end{cases}$, which has the

solution $a = \frac{1}{8}$ and $b = -\frac{1}{16}$. In conclusion, $y = C_1 + C_2 e^{4x} + \left(\frac{1}{8}x^2 - \frac{1}{16}x\right)e^{4x}$. 

 **Example 3.3.4** Determine the general solution of the equation $y^V - 2y^{IV} + y''' - 2y'' = -x + e^x$.

 **Solution:** The solution of the homogeneous equation is $y_0 = C_1 + C_2 x + C_3 e^{2x} + C_4 \sin x + C_5 \cos x$. Because of the linearity of the differential equation, we can determine the particular solution in the form $y_p = y_{p1} + y_{p2}$, where y_{p1} , y_{p2} are particular solutions of the equations $L[y] = f_1$, respectively $L[y] = f_2$ where $f_1 = -x$ and $f_2 = e^x$.

We search for $y_{p1} = x^2(ax + b)$ (since 0 is a double root of the characteristic equation). Substituting in the equation $L[y] = f_1$ we find $a = \frac{1}{12}$, $b = \frac{1}{8}$.

For f_2 we have $y_{p2} = ce^x$, with $c = -\frac{1}{2}$. Finally, $y = C_1 + C_2 x + C_3 e^{2x} + C_4 \sin x + C_5 \cos x + \frac{x^3}{12} + \frac{x^2}{8} - \frac{e^x}{2}$. 

 **Example 3.3.5** Determine the solution for the Cauchy problem: $y''' - 3y'' + 3y' - y = e^x \cos 2x$, $y(0) = 0$, $y'(0) = \frac{1}{4}$, $y''(0) = \frac{1}{2}$.

 **Solution:** The characteristic equation $r^3 - 3r^2 + 3r - 1 = 0$ has the triple root $r = 1$, so $y_0 = (C_1 + C_2x + C_3x^2)e^x$.

We search the particular solution in the form $y_p = e^x(a \cos 2x + b \sin 2x)$. Substituting in the equation and identifying the coefficients we obtain the system:

$$\begin{cases} -11a - 2b - 3(4b - 3a) + 3(a + 2b) - a = 1 \\ -11b + 2a - 3(-4a - 3b) + 3(b - 2a) - b = 0 \end{cases}$$

with the solution $a = 0$, $b = -\frac{1}{8}$. The general solution of the equation is

$y = (C_1 + C_2x + C_3x^2)e^x - \frac{e^x}{8} \sin 2x$. From the initial conditions we get the system:

$$\begin{cases} C_1 = 0 \\ C_1 + C_2 - \frac{1}{4} = \frac{1}{4} \\ C_1 + 2C_2 + 2C_3 - \frac{1}{2} = \frac{1}{2} \end{cases}$$

that gives $C_1 = 0$, $C_2 = \frac{1}{2}$, $C_3 = 0$.

In conclusion, the solution is $y = \frac{e^x}{2} \left(x - \frac{1}{4} \sin 2x \right)$. 



Exercices

 **3.3.2** Determine the general solutions for the following equations:

- | | |
|---|--|
| i). $y''' - 6y'' + 11y' - 6y = -18x + 39$ | vii). $y'' + 4y = \cos 2x$ |
| ii). $2y'' + 5y' - 3y = 5(\sin x + \cos x)$ | viii). $y'' + 6y' + 13y = e^{-x} \cos x$ |
| iii). $y^{IV} + y = x^2$ | ix). $2y'' - 5y' + 2y = x^2 e^x$ |
| iv). $y''' + y' = x - 2$ | x). $y'' - 2y' + y = 2 + e^x \sin x$ |
| v). $y'' - 2y' + y = e^{2x}$ | xi). $y'' + 4y = x \sin^2 x$ |
| vi). $y''' - y'' + y' - y = x^2 + x$ | |

 **3.3.3** In the following problems, indicate the general form of a particular solution, without calculating the coefficients.

- $y'' - 12y' + 35y = xe^{5x} + e^{3x} \cos x + x^2 - x + 2$
- $y''' - 2y'' + 4y' - 8y = \cos 2x + e^x \cos 2x$

$$\text{iii). } y^{IV} - 2y''' + y'' = x^3 + 1$$

$$\text{iv). } y'' - 8y' + 20y = 5xe^{4x} \sin 2x$$

$$\text{v). } y'' + y = \sin x \sin 2x$$

$$\text{vi). } y^V + y^{IV} = xe^{-x} + 3$$

$$\text{vii). } y'' - a^2y = e^{bx}, a, b \in \mathbb{R}$$

 **3.3.4** Determine the solutions for the following Cauchy problems or bi-local conditions problems:

$$\text{i). } y'' + y = \sin x - \cos x, y(\frac{\pi}{2}) = -\frac{\pi}{4}, y'(\frac{\pi}{2}) = -\frac{1}{2}$$

$$\text{ii). } y''' - 3y' - 2y = 9e^{2x}, y(0) = 0, y'(0) = -3, y''(0) = 3$$

$$\text{iii). } y'' + 2y' + 2 = 0, y(0) = e^2, y(1) = 0$$

$$\text{iv). } y'' + 4y = 4x, y(0) = 2, y(\frac{\pi}{4}) = \frac{\pi}{4}$$

$$\text{v). } y'' + 4y' + 3y = 8e^x + 9, \text{ such that } y \rightarrow 3 \text{ when } x \rightarrow -\infty$$

$$\text{vi). } y'' - 4y' + 5y = 2x^2e^x, y(0) = 1, y'(0) = -2$$



3.3.1 i) $y = C_1 \cos x + C_2 \sin x - x \cos x + \sin x \ln |\sin x|$; ii) $y = C_1 + C_2 e^{2x} - \cos e^x - e^x \sin e^x$; iii) $y = C_1 e^x + C_2 x e^x + x e^x \ln |x|$; iv) $y = C_1 + C_2 e^x + (e^x - 1) \ln |e^x - 1| - x(e^x - 1) + 1$; v) $y = C_1 e^x + C_2 e^{-x} + \frac{1}{2}(e^x - e^{-x}) \ln(e^x + 1) - \frac{e^x}{4} + \frac{1}{2}$; vi) $y = C_1 + C_2 \cos x + C_3 \sin x + \frac{1}{\cos x} + \cos x \ln |\cos x| - \tan x \sin x + x \sin x$; vii) $y = C_1 + C_2 x + C_3 e^{-x} - x + x \ln |x|$; viii) $y = e^{-x} \left(C_1 + C_2 x + \frac{4}{3} x^{\frac{5}{2}} \right)$.

3.3.2 i) $y = C_1 e^x + C_2 e^{2x} + C_3 e^{3x} + 3x - 1$; ii) $y = C_1 e^{x/2} + C_2 e^{-3x} - \cos x$; iii) $y = e^{\frac{\sqrt{2}}{2}x} \left(C_1 \cos \frac{\sqrt{2}}{2}x + C_2 \sin \frac{\sqrt{2}}{2}x \right) + e^{-\frac{\sqrt{2}}{2}x} \left(C_3 \cos \frac{\sqrt{2}}{2}x + C_4 \sin \frac{\sqrt{2}}{2}x \right) + x^2$; iv) $y = C_1 + C_2 \cos x + C_3 \sin x + \frac{x^2}{2} - 2x$; v) $y = C_1 e^x + C_2 x e^x + e^{2x}$; vi) $y = C_1 e^x + C_2 \cos x + C_3 \sin x - x^2 - 3x - 1$; vii) $y = C_1 \cos 2x + C_2 \sin 2x + \frac{x}{4} \sin 2x$; viii) $y = e^{-3x} (C_1 \cos 2x + C_2 \sin 2x) + e^{-x} \left(\frac{7}{65} \cos x + \frac{4}{65} \sin x \right)$; ix) $y = C_1 e^{2x} + C_2 e^{\frac{1}{2}x} + e^x(-x^2 + 2x - 6)$; x) $y = 2 + e^x(C_1 + C_2 x - \sin x)$; xi) $y = C_1 \cos 2x + C_2 \sin 2x + \frac{x}{8} \left(1 - \frac{\cos 2x}{4} - \frac{x}{2} \sin 2x \right)$.

3.3.3 i) $y_p = x e^{5x}(Ax + B) + e^{3x}(C \cos x + D \sin x) + E x^2 + F x + G$; ii) $y_p = x(A \cos 2x + B \sin 2x) + e^x(C \cos 2x + D \sin 2x)$; iii) $y_p = x^2(Ax^3 + Bx^2 + Cx + D)$; iv) $y_p = x e^{4x}[(Ax + B) \cos 2x + (Cx + D) \sin 2x]$; v) $y_p = x(A \cos x + B \sin x) + C \cos 3x + D \sin 3x$; vi) $y_p = x(Ax + B)e^{-x} + Cx^4$; vii) $y_p = A e^{bx}$, if $a \neq b$ and $a \neq -b$, $y_p = A x e^{bx}$, if $a = b$ or $a = -b$.

3.3.4 i) $y = \frac{\pi}{4} \cos x - \frac{1}{2}x(\cos x + \sin x)$; ii) $y = (x - 1)(e^{2x} - e^{-x})$; iii) $y = e^{2(1-x)} - x$; iv) $y = x + 2 \cos 2x$; v) $y = e^x + 3$; vi) $y = -5e^{2x} \sin x + (x^2 + 2x + 1)e^x$.

3.4

Linear differential equations that can be reduced to equations with constant coefficients

A linear differential equation of the form

$$(ax+b)^n y^{(n)} + a_1(ax+b)^{n-1} y^{(n-1)} + \cdots + a_{n-1}(ax+b) y' + a_n y = f(x) \quad (3.4.1)$$

where $a_1, \dots, a_n, a, b \in \mathbb{R}$, and f is continuous on a real interval, is called **Euler's equation**.

Performing the change of variable $ax + b = e^t$ we obtain a linear equation with constant coefficients, with the unknown function $y(t)$.

 **Example 3.4.1** Integrate the equation $(x + 1)^2 y'' + 2(x + 1)y' - 12y = 10x + 10 - 12 \ln(x + 1)$.

 **Solution:** We denote $x + 1 = e^t$ and calculate the derivatives of y with respect to x :

$$\begin{aligned} y' &= \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{dy}{dt} e^{-t} \\ y'' &= \frac{dy'}{dx} = \frac{\frac{dy'}{dt}}{\frac{dx}{dt}} = e^{-t} \left(\frac{d^2 y}{dt^2} e^{-t} - \frac{dy}{dt} e^{-t} \right) = e^{-2t} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right) \end{aligned} \quad (3.4.2)$$

Substituting in the initial equation we get

$$\frac{d^2 y}{dt^2} + \frac{dy}{dt} - 12y = 10e^t - 12t.$$

The solution of the homogeneous equation is $y_0 = C_1 e^{3t} + C_2 e^{-4t}$. We search a particular solution of the form $y_p = ae^t + bt + c$ and using the method of unknown coefficients, we obtain $a = -1$, $b = 1$, $c = \frac{1}{12}$, so $y(t) = C_1 e^{3t} + C_2 e^{-4t} - e^t + t + \frac{1}{12}$ and $y(x) = C_1 (x+1)^3 + C_2 (x+1)^{-4} - x + \ln(x+1) - \frac{11}{12}$. 

For the homogeneous Euler equation, which can be reduced to the form

$$x^n y^{(n)} + a_1 x^{n-1} y^{(n-1)} + \dots + a_{n-1} x y' + a_n y = 0 \quad (3.4.3)$$

there exists also a direct method of finding the solution, that is, searching it in the form $y = x^r$. The characteristic equation is:

$$r(r-1)(r-2)\dots(r-n+1) + a_1 r(r-1)\dots(r-n+2) + \dots + a_{n-1} r + a_n = 0$$

1) If r is a real root of order q of the characteristic equation, we have the following q linearly independent solutions:

$$y_1 = x^r, \quad y_2 = x^r \ln x, \quad \dots, \quad y_q = x^r (\ln x)^{q-1}.$$

2) If $\alpha \pm \beta i$ is a pair of complex, conjugated roots of order q for the characteristic equation, $2q$ linearly independent solutions correspond:

$$\begin{aligned} y_1 &= x^\alpha \cos(\beta \ln x), \quad y_2 = x^\alpha \ln x \cos(\beta \ln x), \quad \dots, \quad y_q = x^\alpha (\ln x)^{q-1} \cos(\beta \ln x), \\ y_{q+1} &= x^\alpha \sin(\beta \ln x), \quad y_{q+2} = x^\alpha \ln x \sin(\beta \ln x), \quad \dots, \quad y_{2q} = x^\alpha (\ln x)^{q-1} \sin(\beta \ln x). \end{aligned}$$

 **Example 3.4.2** Determine the general solution of the equation $(2x-3)^2 y'' - 5(2x-3)y' + 10y = 0$.

 **Solution:** We make first the change of variable $2x - 3 = t$, to bring the equation in the form (3.4.3). We have $y' = 2 \frac{dy}{dt}$, $y'' = 4 \frac{d^2y}{dt^2}$ and substituting in the equation, it follows

$$4t^2 \frac{d^2y}{dt^2} - 10t \frac{dy}{dt} + 10 = 0$$

We write the characteristic equation $4r(r-1) - 10r + 10 = 0$, with the roots $r_1 = 1$ and $r_2 = \frac{5}{2}$. So the solution of the equation is $y(x) = C_1(2x-3) + C_2(2x-3)^{\frac{5}{2}}$. 

 **Example 3.4.3** Solve the Cauchy problem: $x^2 y'' - xy' + y = 2x$, $y(1) = 0$, $y'(1) = 1$.

 **Solution:** Making the change of variable $x = e^t$ we obtain $\frac{d^2y}{dt^2} - 2 \frac{dy}{dt} + y = 2e^t$, with the homogeneous equation's solution $y_0 = C_1 e^t + C_2 t e^t$. To determine the particular solution y_p we use the method of unknown coefficients. Looking for $y_p = at^2 e^t$ we obtain $a = 1$. In conclusion, the solution of the initial equation is $y(x) = C_1 x + C_2 x \ln x + x(\ln x)^2$. 



Exercises

 **3.4.1** Solve the following equations or Cauchy problems:

- i). $(x-3)^2 y'' + (x-3)y' + y = 0$
- ii). $x^3 y''' - 2x^2 y'' + 4xy' - 4y = 0$
- iii). $(x+1)^3 y''' - 3(x+1)^2 y'' + 6(x+1)y' - 6y = 0$
- iv). $x^2 y'' + xy' - y = 0$, $y(1) = 1$, $y'(1) = 3$
- v). $(3x)^2 y'' - (3x)y' - 12y = 3 \ln(3x)$
- vi). $x^2 y'' - 2y = \cos(\ln x)$
- vii). $x^2 y'' - xy' + y = \frac{\ln x}{x} + \frac{x}{\ln x}$
- viii). $x^2 y'' - xy' + 10y = x$

ix). $x^2y'' + 5xy' + 3y = \sin x$

x). $x^2y'' - 6y = 10x^3$, $y(1) = 6$, $y'(1) = 0$

xi). $x^2y''' - 3xy'' + 3y' = 0$

xii). $(2x + 1)^2y''' - 2(2x + 1)y'' + 4y' = 0$



Answers

3.4.1 i) $y = C_1 \sin(\ln(x - 3)) + C_2 \cos(\ln(x - 3))$; ii) The characteristic equation is $r(r - 1)(r - 2) - 2r(r - 1) + 4r - 4 = 0$, and the solution $y = C_1x + C_2x^2 + C_3x^2 \ln x$; iii) $y = C_1(x + 1) + C_2(x + 1)^2 + C_3(x + 1)^3$; iv) $y = 2x - \frac{1}{x}$; v) We have $y' = \frac{dy}{dt} 3e^{-t}$, $y'' = \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) 9e^{-2t}$, and the solution is $y = C_1(3x)^2 + C_2(3x)^{-\frac{2}{3}} - \frac{\ln(3x)}{4} + \frac{1}{4}$; vi) $y = \frac{C_1}{x} + C_2x^2 - \frac{3}{10} \cos(\ln x) - \frac{1}{10} \sin(\ln x)$; vii) The method of constants' variation is used, $y = C_1x + C_2x \ln x + \frac{1}{4x}(1 + \ln x) + x \ln x \ln |\ln x|$; viii) $y = x[C_1 \cos(3 \ln x) + C_2 \sin(3 \ln x)] + \frac{x}{9}$; ix) $y = \frac{C_1}{x} + \frac{C_2}{x^3} - \frac{\sin x}{x^2} - \frac{\cos x}{x^3}$; x) $y = \frac{4}{x^2} + 2x^3 = 2x^3 \ln x$; xi) First we make the substitution $y' = z$; the solution is $y = C_1x^4 + C_2x^2 + C_3$; xii) $y = C_1(2x + 1)^2 \left[\ln(2x + 1) - \frac{1}{2} \right] + C_2(x^2 + x) + C_3$.

3.5

Linear differential equations with variable coefficients

As we saw in the previous sections, in order to solve a linear differential equation $L[y] = f(x)$ it is essential to find a fundamental system of solutions for the associated homogeneous equation $L[y] = 0$. For equations with nonconstant coefficients there are no general methods for determining such a system. If a particular solution y_1 of the equation $L[y] = 0$ is known, by making the substitution $y = y_1z$ the order of the equation decreases with one unit.

Example 3.5.1 Solve the equation $(x^2 + 1)y'' - 2xy' + 2y = 0$ knowing that it admits the solution $y_1 = x$.

☞ **Solution:** We make the substitution $y = xz$. We have $y' = z + xz'$ and $y'' = 2z' + xz''$. Substituting in the initial equation we obtain $(x^3 + x)z'' + 2z' = 0$.

Performing now a second change of variable, $z' = u$, we have $(x^3 + x)u' = -2u$, which is an equation with separable variables: $\frac{u'}{u} = -2 \left(\frac{1}{x} - \frac{x}{x^2 + 1} \right)$. From here, integrating, we obtain $\ln u = \ln(x^2 + 1) - \ln x^2 + \ln C_1$, that is $u = \frac{C_1(x^2 + 1)}{x^2}$.

From here $z = C_1x - \frac{C_1}{x} + C_2$ and next $y = C_1(x^2 - 1) + C_2x$. ✓

🔑 **Example 3.5.2** Integrate the equation $xy'' + 2(1 - x)y' + (x - 2)y = 2e^x$, knowing that the corresponding homogeneous equation admits the solution $y_1 = e^x$.

☞ **Solution:** We make the substitution $y = e^x z$. We get $y' = e^x z + e^x z'$ and $y'' = e^x z + 2e^x z' + e^x z''$. Replacing in the equation we obtain $xz'' + 2z' = 0$, that is $\frac{z''}{z'} = -\frac{2}{x}$. Integrating, we have $z' = \frac{C_1}{x^2}$ and next $z = \frac{C_1}{x} + C_2$, so $y_0 = \frac{C_1}{x}e^x + C_2e^x$.

To determine the particular solution $y_p = \varphi_1(x)\frac{e^x}{x} + \varphi_2(x)e^x$ by the method of constants variation, we write the nonhomogeneous equation in the form (3.1.1) $y'' + \frac{2(1-x)}{x}y' + \frac{x-2}{x}y = \frac{2e^x}{x}$. The functions φ_1 and φ_2 are obtained from the system:

$$\begin{cases} \varphi_1' \frac{e^x}{x} + \varphi_2' e^x = 0 \\ \varphi_1' \left(\frac{e^x}{x} - \frac{e^x}{x^2} \right) + \varphi_2' e^x = \frac{2e^x}{x} \end{cases}.$$

We have $\varphi_1' = -2x$, $\varphi_2' = 2$, so $\varphi_1 = -x^2$, $\varphi_2 = 2x$ and $y_p = xe^x$.

In conclusion, the general solution is $y(x) = \left(\frac{C_1}{x} + C_2 + x \right) e^x$. ✓

In the case of a homogeneous equation of order two, of the form

$$y'' + a_1(x)y' + a_2(x)y = 0$$

for which a particular solution y_1 is known, a second particular solution y_2 , linearly independent with respect to the first one, can be determined using the Ostrogradski-Liouville formula:

$$\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = C_1 e^{-\int a_1(x) dx} \quad (3.5.1)$$

 **Example 3.5.3** Integrate the linear differential equation $xy'' + 2y' + xy = 0$, knowing that it admits the solution $y_1 = \frac{\sin x}{x}$.

 **Solution:** We apply formula (3.5.1) for the equation $y'' + \frac{2}{x}y' + y = 0$. If y_2 is another particular solution, we have $\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = C_1 e^{-\int \frac{2}{x}} = C_1 e^{-2 \ln x} = \frac{C_1}{x^2}$, that is $y_1 y_2' - y_1' y_2 = \frac{C_1}{x^2}$. We write this in the form $\left(\frac{y_2}{y_1}\right)' y_1^2 = \frac{C_1}{x^2}$ and we replace the given y_1 . It follows $\left(y_2 \frac{x}{\sin x}\right)' = \frac{C_1}{\sin^2 x}$. From here, by integration, we have $y_2 \frac{x}{\sin x} = -\frac{C_1}{\tan x} + C_2$. In conclusion, $y_2 = \frac{C_1 \cos x + C_2 \sin x}{x}$, so the general solution will be $y = C_1 \frac{\cos x}{x} + C_2 \frac{\sin x}{x}$. 



Exercises

 **3.5.1** Integrate the following linear homogeneous equations, knowing that they admit the indicated particular solutions:

- i). $y'' - 2(1 + \tan^2 x)y = 0$, $y_1 = \tan x$
- ii). $y'' + 2 \cdot \frac{\cos x}{\sin x} y' + 3y = 0$, $y_1 = \cos x$
- iii). $(1 + x^2)y'' + xy' - y = 0$, $y_1 = x$
- iv). $(2x + 1)y'' + (4x - 2)y' - 8y = 0$, $y_1 = e^{-2x}$
- v). $x^2 y'' - xy' - 8y = 0$, $y_1 = \frac{1}{x^2}$
- vi). $y'' - y' + ye^{2x} = 0$, $y_1 = \sin(e^x)$
- vii). $(2x + 1)y'' + 4xy' - 4y = 0$, y_1 is of the form e^{ax}
- viii). $(x - 1)y'' - xy' + y = 0$, y_1 is of the form e^{ax}
- ix). $x(x - 1)y'' - (2x - 1)y' + 2y = 0$, y_1 is a polynomial of degree 1
- x). $xy''' - y'' - xy' + y = 0$, $y_1 = e^x$, $y_2 = x$
- xi). $x^3 y''' - 3x^2 y'' + 6xy' - 6y = 0$, $y_1 = x$, $y_2 = x^2$

 **3.5.2** Integrate the following nonhomogeneous equations, knowing that the homogeneous equations admit the indicated particular solutions:

i). $x^2(1 - \ln x)y'' + xy' - y = \frac{(1 - \ln x)^2}{x}$, $y_1 = \ln x$, $y_2 = x$

ii). $xy'' - (2x + 1)y' + (x + 1)y = 2x^2e^x$, $y_1 = x^2e^x$, $y_2 = e^x$

iii). $4xy'' + 2y' + y = 1$, $y_1 = \sin \sqrt{x}$, $y_2 = \cos \sqrt{x}$

iv). $xy'' + 2y' - xy = e^x$, $y_1 = \frac{e^x}{x}$

v). $(x + 1)xy'' + (x + 2)y' - y = x + \frac{1}{x}$, y_1 polynomial of degree 1

 **3.5.3** Find the general solution of the equation $(x^2 - 1)y'' + 4xy' + 2y = 6x$, knowing that $y_1 = x$, $y_2 = \frac{x^2 + x + 1}{x + 1}$ are two particular solutions of the nonhomogeneous equation.

 **3.5.4** Making a suitable change of variable, reduce the following equations to equations with constant coefficients:

i). $(x^2 + 1)^2y'' + 2x(x^2 + 1)y' - y = 0$

ii). $x^2y'' + 4xy' + (2 - x^2)y = 1$



Answers

3.5.1 i) $y = C_1(1 + x \tan x) + C_2 \tan x$; ii) $y = C_1 \left(\frac{\cos^2 x}{\sin x} - \sin x \right) + C_2 \cos x$; iii) $y = C_1x + C_2\sqrt{1 + x^2}$; iv) $y = C_1(1 + 4x^2) + C_2e^{-2x}$; v) $y = \frac{C_1}{x^2} + C_2x^4$; vi) $y = C_1 \cos e^x + C_2 \sin e^x$; vii) $y = C_1x + C_2e^{-2x}$; viii) $y = C_1x + C_2e^x$; ix) $y = C_1(2x - 1) + C_2x^2$; x) Using the substitution $y = y_1z$ the equation $xu'' + (3x - 1)u' + 2(x - 1)u = 0$ is obtained, where $u = z' = \left(\frac{y}{y_1} \right)'$.

This equation has the particular solution $u_1 = \left(\frac{y_2}{y_1} \right)' = \left(\frac{x}{e^x} \right)'$. Making now the substitution $u = u_1t$, a first order equation with the unknown t will follow. Finally we obtain $y = C_1x + C_2e^x + C_3e^{-x}$; xi) $y = C_1x + C_2x^2 + C_3x^3$.

3.5.2 i) The method of constants' variation is used, $y = C_1 \ln x + C_2x + \frac{1 - 2 \ln x}{4x}$; ii) $y = \left(C_1x^2 + C_2 + \frac{2}{3}x^3 \right) e^x$; iii) Searching a particular solution of the form $y_p = \phi_1(x) \sin \sqrt{x} + \phi_2(x) \cos \sqrt{x}$, with the method of constants

variation, we find $y_p = 1$, so $y = C_1 \sin \sqrt{x} + C_2 \cos \sqrt{x} + 1$; **iv)** $y = C_1 \frac{e^x}{x} + C_2 \frac{e^{-x}}{x} + \frac{e^x}{4} \left(2 - \frac{1}{x} \right)$; **v)** $y = C_1(x+2) + \frac{C_2}{x} + \left(\frac{x}{2} + 1 \right) \ln |x| + \frac{3}{2}$.

3.5.3 $y_2 - y_1 = (x+1)^{-1}$ is a solution of the associated homogeneous equation. We obtain $y = C_1(x+1)^{-1} + C_2(x-1)^{-1} + x$.

3.5.4 **i)** $x = \tan t$, $y''_{t^2} - y = 0$; **ii)** $z = x^2 y$, $z'' - z = 0$.